LINUX SERVER PRACTICAL

BIND DNS Server Configuration

Questions with Answers • Step-by-step practical guide

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Task Name	Install and Configure BIND DNS with A and CNAME Records
Domain for Lab	myzomatoclone.local
DNS Server IP	127.0.0.1 (loopback / local machine)

Important: The task mentions the Debian/Ubuntu BIND configuration path `/etc/bind/`. Since RHEL uses different paths, this guide includes both sets of commands. Follow only the section matching your Linux distribution.

Platform paths at a glance

Distribution	Package / service	Main configuration	Zone files
Ubuntu / Debian	bind9 / named	/etc/bind/named.conf.local	/etc/bind/
RHEL 9 / Rocky / Alma	bind / named	/etc/named.conf	/var/named/

Task 1 — Install BIND and verify the version

Question: Install the BIND DNS server and verify the installation with the named -v command.

Answer: Install the package for your distribution, then check the installed version.

Ubuntu / Debian

```
sudo apt update
sudo apt install bind9 bind9-utils dnsutils
named -v
```

RHEL 9 / Rocky / AlmaLinux

```
sudo dnf install bind bind-utils
named -v
```

Expected result: the command prints a BIND version, for example a line beginning with `BIND`. The exact version depends on the repository and updates installed.

Check service status

```
sudo systemctl status named
# On Ubuntu/Debian, this may also be called:
sudo systemctl status bind9
```

Meaning: BIND is the DNS server software. The service listens for DNS queries and answers using its configured zones.

Task 2 — Create a forward DNS zone

Question: Configure a zone for myzomatoclone.local and map the domain to 127.0.0.1. Restart the DNS service after making the changes.

Answer: Create a zone declaration and a zone file with an SOA record and an A record.

Option A: Ubuntu / Debian

1. Edit the local zone configuration:

```
sudo nano /etc/bind/named.conf.local
```

Add this at the end of the file:

```
zone "myzomatoclone.local" {
    type master;
    file "/etc/bind/db.myzomatoclone.local";
};
```

2. Create the zone file:

```
sudo cp /etc/bind/db.local /etc/bind/db.myzomatoclone.local
sudo nano /etc/bind/db.myzomatoclone.local
```

Replace the file contents with:

```
$TTL 86400
@   IN  SOA localhost. admin.myzomatoclone.local. (
        2026101901 ; serial (increase after each edit)
        3600       ; refresh
        1800       ; retry
        604800     ; expire
        86400 )    ; negative cache TTL

@           IN  NS      localhost.
@           IN  A       127.0.0.1
```

3. Validate the configuration and zone, then restart BIND:

```
sudo named-checkconf
sudo named-checkzone myzomatoclone.local /etc/bind/db.myzomatoclone.local
sudo systemctl restart bind9
sudo systemctl enable bind9
```

Option B: RHEL 9 (use this if your lab machine is RHEL)

1. Add a zone declaration to `/etc/named.conf`:

```
zone "myzomatoclone.local" IN {
    type master;
    file "myzomatoclone.local.zone";
    allow-update { none; };
};
```

2. Create the zone file. RHEL's standard zone directory is `/var/named`.

```
sudo nano /var/named/myzomatoclone.local.zone
```

```
$TTL 86400
@   IN  SOA ns1.myzomatoclone.local. admin.myzomatoclone.local. (
        2026101901
        3600
        1800
        604800
        86400 )

@       IN  NS  ns1.myzomatoclone.local.
ns1     IN  A   127.0.0.1
@       IN  A   127.0.0.1
```

3. Validate and restart:

```
sudo named-checkconf
sudo named-checkzone myzomatoclone.local /var/named/myzomatoclone.local.zone
sudo systemctl enable --now named
sudo systemctl restart named
```

Note: For a real LAN DNS server, the nameserver A record and domain A record should normally point to the server's reachable LAN IP, not 127.0.0.1. Here 127.0.0.1 is used because the assignment specifically requests it.

Task 3 — Use the local resolver and test the domain

Question: Set the system's DNS resolver to the local BIND server and test myzomatoclone.local with dig or nslookup.

Answer: First test BIND directly. This confirms the zone works even before changing system-wide resolver settings.

Direct test (works on both distributions)

```
dig @127.0.0.1 myzomatoclone.local A
dig @127.0.0.1 myzomatoclone.local NS
nslookup myzomatoclone.local 127.0.0.1
```

Expected result: the A query should return `127.0.0.1` and usually a successful response status such as `NOERROR`.

Set the resolver

Ubuntu / Debian: if systemd-resolved is managing DNS, use its supported configuration rather than permanently editing a generated `/etc/resolv.conf`. For a simple lab, you can test with `dig @127.0.0.1` first. To configure systemd-resolved, edit `/etc/systemd/resolved.conf` and set `DNS=127.0.0.1`, then restart it:

```
sudo nano /etc/systemd/resolved.conf
# Under [Resolve], set:
DNS=127.0.0.1

sudo systemctl restart systemd-resolved
resolvectl status
```

RHEL 9: set the connection's DNS server with NetworkManager. Replace `ens160` if your connection profile has a different name:

```
nmcli connection show
sudo nmcli connection modify ens160 ipv4.dns 127.0.0.1
sudo nmcli connection modify ens160 ipv4.ignore-auto-dns yes
sudo nmcli connection up ens160
cat /etc/resolv.conf
```

Caution: setting 127.0.0.1 as system DNS will make the machine depend on its local BIND service for name lookups. If DNS stops working, use the direct dig test and restore your previous DNS settings if needed. On managed networks, do not overwrite existing DNS settings without permission.

Task 4 — Add and verify a CNAME alias

Question: Add foodsearch.myzomatoclone.local as a CNAME pointing to myzomatoclone.local, then verify it with dig.

Answer: A CNAME record creates an alias for another DNS name. Add it as a separate record in the same zone file.

Add this line in the zone file, below the existing A record:

```
foodsearch  IN  CNAME  myzomatoclone.local.
```

For Ubuntu/Debian, edit `/etc/bind/db.myzomatoclone.local`. For RHEL, edit `/var/named/myzomatoclone.local.zone`. Increase the SOA serial number after every zone-file change (for example, from 2026101901 to 2026101902). Then validate and reload:

```
# Ubuntu / Debian
sudo named-checkzone myzomatoclone.local /etc/bind/db.myzomatoclone.local
sudo systemctl reload bind9

# RHEL 9
sudo named-checkzone myzomatoclone.local /var/named/myzomatoclone.local.zone
sudo systemctl reload named
```

Verify the alias:

```
dig @127.0.0.1 foodsearch.myzomatoclone.local CNAME
```

```
dig @127.0.0.1 foodsearch.myzomatoclone.local A
```

The CNAME query should show `foodsearch.myzomatoclone.local.` pointing to `myzomatoclone.local.`. The A query should follow the alias and ultimately resolve to 127.0.0.1.

Task 5 — Fix the incorrect zone-file snippet

Question: Find and fix the error in: 'myzomatoclone.local. IN A 127.0.0.1 foodsearch IN CNAME myzomatoclone.local.' The CNAME is not resolving as expected.

Answer: The two DNS records have been written on one line without a record boundary. Put each record on its own line, and use the relative owner name foodsearch inside this zone.

Incorrect:

```
myzomatoclone.local. IN A 127.0.0.1 foodsearch IN CNAME myzomatoclone.local.
```

Correct (inside the myzomatoclone.local zone file):

```
@           IN  A       127.0.0.1
foodsearch  IN  CNAME   myzomatoclone.local.
```

Why this works:

- @ means the zone's root name, myzomatoclone.local.
- `foodsearch` expands to foodsearch.myzomatoclone.local.
- A CNAME must be a separate resource-record line.
- The target name ends with a dot so it is treated as a fully qualified domain name.
- A CNAME owner should not also have other record types (apart from DNSSEC-related records).

After fixing it, increment the SOA serial, run the named-checkzone command for your distribution, reload BIND, and run the Task 4 dig commands again.

Troubleshooting checklist

Problem	What to check
BIND will not start	Run named-checkconf; inspect systemctl status named or bind9.
Zone fails validation	Check SOA parentheses, dots, record layout, and file path.
Old DNS answer appears	Increase SOA serial and reload/restart the service.
dig @127.0.0.1 times out	Check service status, listen-on settings, and firewall rules for DNS (UDP/TCP port 53).
.local behaves unexpectedly	.local is commonly used by mDNS. For general labs, a reserved test domain such as myzomatoclone.test is le

Conclusion: BIND was installed and version-checked; a forward zone and A record were configured; local resolution was tested; a CNAME alias was added; and the malformed record syntax was corrected.